

ADITYA MITTAL

SOFTWARE ENGINEER BUILDING PRODUCTION AI-INTEGRATED SYSTEMS — LLM PROMPTING PIPELINES WITH SAFETY CONSTRAINTS, RESTFUL APIS, AND STATIC-ANALYSIS TOOLING IN DJANGO, PYTHON, AND REACT.

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EDUCATION

Jaypee University of Information Technology

2023 – 2027

B.Tech – Computer Science and Engineering (Minor: ECE)

Solan, Himachal Pradesh

EXPERIENCE

Basepair

June 2026 – August 2026

Software Development Engineer in Test (SDET) Intern

Remote

- Diagnosed and resolved critical production distributed-systems defects, including an OAuth race condition via DB-level row locking and multi-instance file-access failures via AWS S3 on-demand fallbacks.
- Led a full-stack SonarQube remediation initiative across Python/Django and React codebases, resolving 100+ BLOCKER/HIGH security and reliability defects (SQL injection, ReDoS, insecure crypto, unhandled exceptions) and authoring organizational Standards.md guidelines.

Spelll Production

June 2024 – July 2024

Frontend Engineering Intern

Remote

- Owned the end-to-end frontend lifecycle from prototyping to production deployment, translating UI/UX mockups into pixel-perfect, mobile-first components and integrating client-side form validation to improve engagement and lead conversion.

PROJECTS

MedVault | Django, Django REST Framework, PostgreSQL, Google Gemini API, React, TypeScript, JWT, Docker | [LIVE](#)

Feb 2026 – August 2026

- Engineered a structured Google Gemini API prompting pipeline with a hard-coded JSON output schema and explicit safety constraints preventing diagnosis/prescription hallucinations, capping input at 50,000 characters and adding a tolerant JSON-from-prose fallback parser for malformed model output.
- Designed a Django REST Framework backend with a custom email-based User model, JWT authentication with refresh-token rotation, and per-user data isolation enforced at the queryset level; isolated all Gemini calls behind a single mockable client so a 47-test backend suite runs with zero API key configured.

Canopy | Django, PostgreSQL, tree-sitter, Python ast, Google Gemini API, Plotly Dash, Docker | [LIVE](#)

August 2026 – September 2026

- Designed a multi-language static-analysis engine unifying Python's ast module and tree-sitter behind one generic analyzer interface, supporting 10 languages with deterministic cross-file call-graph resolution.
- Architected an optional AI chat feature fully isolated from the deterministic analysis pipeline — grounded only in already-computed call-graph results, keeping the core parsing product 100% LLM-free with zero hallucination risk — backed by a commit-hash-keyed PostgreSQL cache and least-privilege, zero-scope GitHub OAuth.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, C, C++, SQL, HTML5, CSS3/SCSS

Frontend: React.js, Next.js, shadCN, Tailwind CSS, EJS

Backend & Frameworks: Django, Django REST Framework, Node.js, Express.js, Temporal, JWT, OAuth 2.0

Cloud & Infrastructure: AWS (S3, STS, EC2, Lambda, RDS, IAM, CloudWatch, EKS, ECS, CloudFront, SQS, SNS, DynamoDB, API Gateway, VPC, ELB, CloudFormation, CDK, Secrets Manager, Parameter Store), Kubernetes, Docker, Vercel

Databases: PostgreSQL, MongoDB

Testing & Quality Assurance: Playwright (E2E Testing), Sentry, Visual Regression Testing, Test Automation Architecture

DevOps & Tooling: Docker, Docker Compose, Terraform, Git, GitHub, npm Workspaces, Postman, Linux/Bash, CI/CD

EXTRACURRICULAR

Technical Sciences and Movies Photography Club (TMP), Member (2023–2025) — organized tech events, coding contests, workshops; mentored peers.

Omega LEO Club, Member (2024–2025) — led community service initiatives.